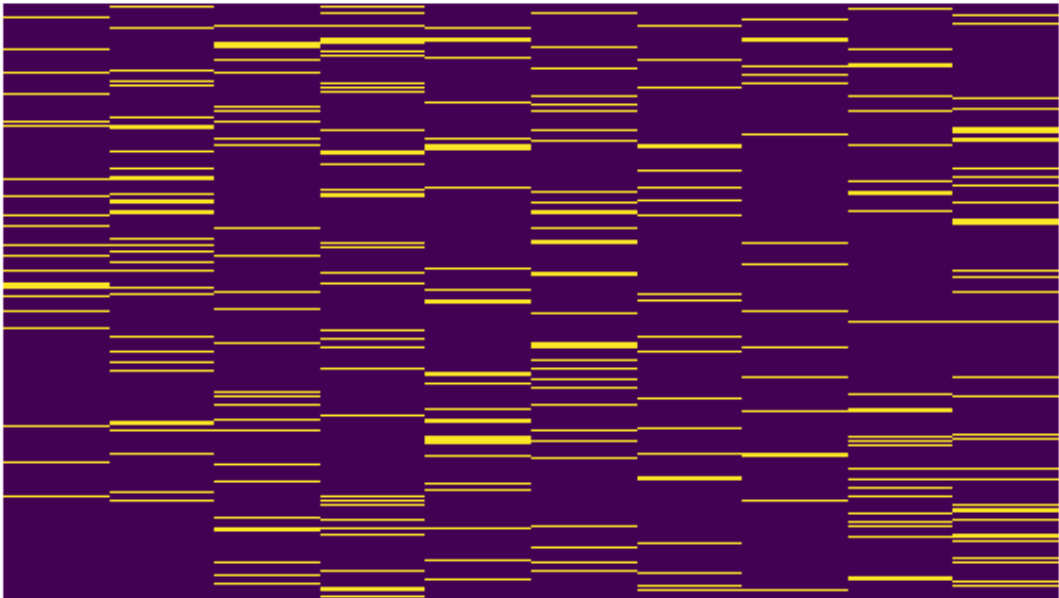
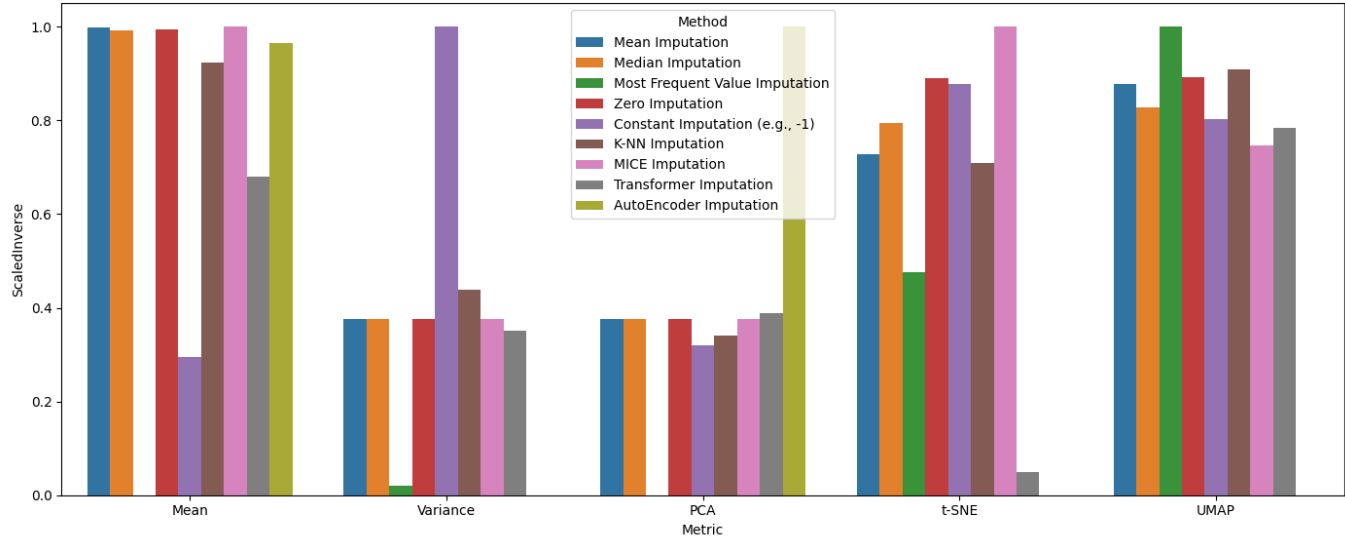


Imputation Evaluation Report

Missing Values Heatmap



Scaled Inverse Scores Across Evaluation Metrics (Higher is Better)



Imputation Evaluation Report

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Method: Mean Imputation

Mean Deviation: 0.0384
Variance Deviation: 0.1484
PCA Distance: 1.8566
t-SNE Distance: 13.0171
UMAP Distance: 2.8351

Method: Median Imputation

Mean Deviation: 0.0387
Variance Deviation: 0.1483
PCA Distance: 1.8565
t-SNE Distance: 12.8383
UMAP Distance: 2.8512

Method: Most Frequent Value Imputation

Mean Deviation: 0.3223
Variance Deviation: 0.7824
PCA Distance: 2.2434
t-SNE Distance: 13.7274
UMAP Distance: 2.7958

Method: Zero Imputation

Mean Deviation: 0.0386
Variance Deviation: 0.1483
PCA Distance: 1.8568
t-SNE Distance: 12.5979
UMAP Distance: 2.8300

Method: Constant Imputation (e.g., -1)

Mean Deviation: 0.1011
Variance Deviation: 0.0612
PCA Distance: 1.9061
t-SNE Distance: 12.6300
UMAP Distance: 2.8596

Method: K-NN Imputation

Mean Deviation: 0.0412
Variance Deviation: 0.1297
PCA Distance: 1.8874
t-SNE Distance: 13.0673
UMAP Distance: 2.8244

Method: MICE Imputation

Mean Deviation: 0.0384
Variance Deviation: 0.1484
PCA Distance: 1.8573
t-SNE Distance: 12.3282
UMAP Distance: 2.8782

Method: Transformer Imputation

Mean Deviation: 0.0535
Variance Deviation: 0.1571
PCA Distance: 1.8465
t-SNE Distance: 15.1196
UMAP Distance: 2.8656

Method: AutoEncoder Imputation

Mean Deviation: 0.0396
Variance Deviation: 1.0432
PCA Distance: 1.4444
t-SNE Distance: 15.3030
UMAP Distance: 3.1515